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Mechanical Engineering
Indian Institute of Technology Bombay

22B2150
B.Tech.
Gender: Male
DOB: 05/11/2004

Examination	University	Institute	Year	CPI / % Credits
Graduation	IIT Bombay	IIT Bombay	2026	C

Pursuing a Minor degree in **Data Science** and **Artificial Intelligence** at **C-MInDS**, IIT Bombay

Professional Experience

Data Scientist | Verint Systems | CX Automation [May'25 - Jul'25]

A global leader in AI-powered customer engagement solutions, serving many of the world's most iconic brands

- Developed a **hybrid B2B recommender** with **LightFM** to suggest enterprise applications and drive **revenue growth**
- Optimized the **BPR loss function** with a **custom negative sampling** strategy weighted by proprietary health scores
- Engineered time-aware features using a **temporal decay function** to align recent user interactions with ARR trends
- Achieved **80% Hit Rate@5** on sparse data using **leave-one-out cross-validation** and hyperparameter optimization

Software Developer | Sherpa Labs | Stanford [Apr'24 - Jul'24]

An **AI-powered oral conversation tool** that provides a better learning experience and improved plagiarism checks

- Developed a **Learning Management System (LMS)** centered on an **AI-powered** oral conversation learning assistant
- Engineered a professional **admin dashboard** for university-level management, implementing a cyber-secure and efficient system with **multiple roles** (admin, super-admin) for centralized control over classes, teachers, and resources
- Utilized **FastAPI's** asynchronous features and **MongoDB** to build a scalable backend with models for data validation

Web Developer Intern | MapIT.ai [Dec'23 - Jan'24]

Built a comprehensive Inventory Management System to streamline operations for all Tinkerer Laboratories nationwide

- Engineered the core backend for a robust Inventory Management System using **Django REST Framework**, developing a suite of **RESTful APIs** for all inventory actions while ensuring code maintainability via thorough documentation
- Utilized **Docker containerization** to encapsulate the codebase, ensuring portability and consistency across environments
- Integrated **SimpleJWT** for stateless user authentication by utilizing **JSON Web Tokens (JWTs)** for enhanced security

Key Projects

Technical General Championship Portal | Web Team | Institute Technical Council [Jun'23 - Aug'23]

- Engineered the Tech GC portal's **Django backend** to manage competition data, submission links, and hostel rankings
- Developed robust **REST APIs** using structured django models and function-based views for efficient data management
- Strengthened portal security with **token-based authentication** to secure all API endpoints for robust data interaction

Movie Recommender System using LightGCN | Course Project: DS303 | Prof. Manjesh Hanawal [Mar'25 - Apr'25]

- Engineered a **LightGCN (Graph Convolutional Network)** recommender on a user-item graph, proving more **efficient** and **scalable** than standard GCNs by removing costly **feature transformations** and surpassing traditional methods
- Effectively mitigated **data sparsity** by uncovering **higher-order relationships** via multi-hop embedding propagation
- Generated **implicit feedback** by treating ratings above a 3-star threshold as positive signals for graph construction

Using Deep RL to Optimize Stock Returns | FinSearch | Finance Club | UGAC [Jun'24 - Jul'24]

- Engineered a **Deep Q-Network** on 10 years of **Nifty 50** data, using indicators like RSI & MACD for **state representation**
- Implemented a custom **Sharpe ratio** reward function and an **epsilon-greedy** strategy to optimize trading policy
- Achieved **18% annualized return** in backtesting, **outperforming** buy-and-hold and standard ARIMA/LSTM models

Facial Emotion Detection using CNN | Course Project: ME228 | Prof. Shyamprasad Karagadde [Mar'24 - Apr'24]

- Designed and developed a **Convolutional Neural Network(CNN)**-based emotion detection model with **73%** accuracy, utilizing **Keras** and **TensorFlow**, employing advanced data preprocessing and augmentation techniques
- Successfully implemented **real-time** emotion recognition using **Haar cascades** and **OpenCV** for live video feed

Position of Responsibility

Web Convener | Institute Technical Council [Jun'23 - Apr'24]

The council responsible for overseeing all technical activities at the institute, catering to **10,000+** students

- Worked within a **5-member** team tasked with developing and maintaining the websites for ITC's clubs and teams
- Represented Web Team in front of **700+** freshers in Freshman ITC Orientation program explaining our team's role
- Contributed to the official ITC portal UI revamp, implementing dynamic animations with **ReactJS** and **React Hooks**

Extra-Curricular Activities

- Actively participated in **Algorithmic Trading** Workshops conducted by Finance Club, IITB, acquiring valuable insights[2024]
- Completed an intensive 8 week-long **DSA Bootcamp** organized by Web and Coding Club, IITB and Coding Ninjas [2024]
- Achieved the **runner-up** spot in Volleyball tournament held during **Freshiesta, Freshman Sports Championship** [2023]
- Secured **1st place** in Shotput event, surpassing over 30 participants, during the **Annual School Sports Meet** [2019]